

ULLEUNGDO, Korea, Republic of

WMO: 471150

Lat: 37.4814N

Lon: 130.8986E

Elev: 224

StdP: 98.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-3.7	-12.6	1.3	-1.6	-10.7	1.5	-1.1	11.9	0.5	10.5	1.4	3.8	20	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.3	29.7	24.4	28.5	24.0	27.4	23.3	25.5	28.3	24.7	27.4	24.1	26.6	2.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.2	27.3	24.0	19.3	26.6	23.3	18.5	25.9	79.6	28.3	76.7	27.5	73.9	26.7	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	9.6	8.4	-7.7	31.9	2.0	1.7	-9.1	33.1	-10.3	34.1	-11.4	35.0	-12.8	36.3
			-8.6	26.5	1.8	1.0	-9.9	27.3	-11.0	27.9	-12.0	28.4	-13.4	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	1.7	2.6	6.0	11.3	16.0	19.3	23.0	24.4	20.3	15.8	10.0	4.3
	DBStd	8.31	3.14	3.46	3.63	3.54	3.05	2.56	2.84	2.60	2.28	2.74	3.73	3.68
	HDD10.0	849	258	208	131	26	1	0	0	0	0	2	46	178
	HDD18.3	2429	517	441	382	212	84	19	2	0	7	83	248	434
	CDD10.0	1926	0	1	8	65	187	278	402	445	309	182	47	2
	CDD18.3	464	0	0	0	1	12	47	146	187	66	6	0	0
	CDH23.3	2257	0	0	0	1	31	101	796	1194	134	1	0	0
	CDH26.7	402	0	0	0	0	3	7	142	241	9	0	0	0
Wind	WSAvg	3.6	3.5	3.7	4.1	4.3	3.9	3.1	3.6	3.3	3.3	3.5	3.7	3.5
Precipitation	PrecAvg	1369	124	86	73	90	96	106	147	157	156	95	113	118
	PrecMax	2239	346	238	180	396	324	308	526	623	467	261	429	309
	PrecMin	770	18	23	8	19	20	18	21	23	13	4	23	25
	PrecStd	335	63	40	38	62	60	63	95	115	99	63	68	64
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	12.9	17.2	22.4	26.4	27.6	30.8	31.9	27.9	22.8	19.4	14.2
		MCWB	7.8	10.1	10.6	13.9	15.7	20.3	24.4	25.5	24.1	18.6	15.1	11.2
	2%	DB	8.5	10.6	14.5	19.8	23.8	25.6	29.5	30.2	26.0	21.5	17.7	12.0
		MCWB	5.5	7.6	9.4	12.6	15.5	19.7	24.0	25.0	22.5	17.8	14.1	9.0
	5%	DB	7.3	8.9	12.7	18.1	22.3	24.4	28.3	29.1	24.8	20.5	16.4	10.7
		MCWB	4.4	6.1	8.4	12.3	15.4	19.3	23.6	24.6	21.5	17.0	13.0	7.6
	10%	DB	6.0	7.4	11.3	16.6	20.8	23.2	27.2	28.0	23.8	19.4	15.1	9.4
		MCWB	3.2	4.7	7.9	11.6	14.8	18.9	23.1	24.0	20.8	16.2	11.9	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	11.1	12.2	16.7	19.7	22.8	26.0	27.0	24.9	20.5	16.8	11.7
		MCDB	9.9	12.4	15.0	19.4	23.1	25.4	28.9	30.2	27.1	21.8	18.2	13.1
	2%	WB	6.1	8.4	10.8	14.7	17.9	21.5	24.9	25.9	23.5	18.9	14.8	9.6
		MCDB	8.1	10.0	12.8	17.9	21.1	23.8	28.1	28.7	25.2	20.6	17.0	11.6
	5%	WB	4.7	6.3	9.5	13.4	16.9	20.6	24.2	25.2	22.4	17.7	13.5	8.0
		MCDB	6.8	8.5	12.1	16.5	20.3	22.8	27.2	27.8	24.1	19.7	15.9	10.5
	10%	WB	3.5	4.9	8.2	12.3	15.9	19.8	23.5	24.6	21.4	16.7	12.1	6.5
		MCDB	5.7	7.3	10.9	15.2	19.1	22.1	26.4	27.1	23.2	18.9	14.8	9.1
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	5.6	6.2	6.0	5.0	4.1	4.3	4.5	4.8	4.8	4.3
		MCDBR	5.5	6.6	7.3	8.0	7.7	6.6	5.5	5.6	5.6	5.8	5.6	
		MCWBR	4.8	5.6	5.9	5.4	5.5	4.5	3.6	3.8	4.4	4.6	4.9	5.0
	5% WB	MCDBR	5.3	6.4	6.8	6.8	6.8	5.5	5.1	5.1	4.8	5.1	5.7	5.6
		MCWBR	5.1	5.8	6.2	5.5	5.7	4.3	3.7	3.7	4.3	4.6	5.3	5.5
Clear-Sky Solar Irradiance	taub	0.409	0.447	0.514	0.539	0.542	0.560	0.551	0.502	0.441	0.435	0.418	0.405	
	taud	2.059	1.985	1.841	1.835	1.891	1.927	2.027	2.178	2.285	2.151	2.118	2.074	
	Ebn at Noon	733	758	743	755	762	746	750	780	809	764	721	704	
	Edh at Noon	124	152	192	205	197	190	171	144	122	127	116	115	
All-Sky Solar Radiation	RadAvg	1.82	2.71	4.04	5.11	5.55	5.50	5.00	5.03	4.14	3.39	2.24	1.65	
	RadStd	0.16	0.24	0.30	0.41	0.60	0.46	0.58	0.47	0.39	0.29	0.23	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	-1.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)		N/A	N/A	-1.23	N/A	N/A	N/A	N/A	N/A	+77	N/A

Nomenclature: See separate page